HR FLOW - ENTERPRISE AI DOCUMENTATION & SOP

How to Add a New HR Department & Train Its 2-Stage AI Model (English + Hinglish)

1. EXECUTIVE OVERVIEW: 2-STAGE AI CLASSIFICATION ARCHITECTURE

Whenever an employee sends an email to the HR Inbox, HR Flow processes the email through a two-stage machine learning and rule pipeline matching the exact architecture of LEAVE and ATTENDANCE:

Stage 1 - Department Detection Router (backend/classification/category_classifier.py):
Scans the email Subject and Body against each department Strong Multi-Word Phrases (keywords_strong) and Weak Single-Word Keywords (keywords_weak) stored in backend/data/categories.json.

* **Strong Multi-Word Phrase Match: +3.2 points in Subject | +2.2 points in Body**
* **Weak Single-Word Keyword Match: +1.5 points in Subject | +0.8 points in Body**
* **Minimum Routing Threshold: >= 1.5 points routes the email to that HR Department.**

Stage 2 - Specialized Intent Model (backend/classification/intent_model_service.py):
Once Stage 1 detects the department (e.g., LEAVE, ATTENDANCE, PAYROLL, BGV, RESIGNATION, or any newly created department), Stage 2 executes that department dedicated scikit-learn model files stored in backend/classification/models/<category_lower>/:

* **<dept>_tfidf_vectorizer.pkl : TfidfVectorizer(lowercase=True, ngram_range=(1, 2), min_df=1, max_df=0.95, sublinear_tf=True)**
* **<dept>_linear_svm.pkl : LinearSVC(C=1.0, random_state=42)**
* **<dept>_intents.json : Maps 8 Intent Codes (e.g., PA01..PA08) to Intent Names**
* **training_dataset.json : Stores the 400 English + Hinglish labeled email samples**
* **model_metadata.json : Stores version, accuracy %, TF-IDF feature count, and timestamp**

2. STEP-BY-STEP GUIDE FOR HR ADMINISTRATORS

STEP 1: Create the New HR Department Desk
1. Open HR Departments from the left sidebar (http://127.0.0.1:5000/inbox.html#departments).
2. Click + Add New HR Department in the top-right header.
3. Enter Department Name, Shared Gmail, SLA Target Hours, Default Priority, and Description. Notice Department Code and Primary Category Code auto-populate.

STEP 2: Configure Stage 1 Department Detection Keywords (Via ChatGPT)
1. Inside the Create Department form (or inside any card Train AI drawer under Stage 1), click Copy ChatGPT Prompt for Department Keywords.
2. Paste the prompt into ChatGPT. It generates 20-28 Strong Multi-Word Phrases and 15-22 Weak Single-Word Keywords covering both English and Indian workplace Hinglish.
3. Paste those comma-separated lists into Strong Multi-Word Phrases and Weak Single-Word Keywords and click Save / Create Department.

HR FLOW - STAGE 2 INTENT TRAINING & JSON SCHEMA

Training 400 English + Hinglish Samples (.pkl + TF-IDF) & Live Testing

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STEP 3: Train Stage 2 Intent Identification Model (400 English + Hinglish Samples)

1. On the department card, click Train AI to open the AI Department Detection & Intent Trainer Studio.
2. Under Stage 2 - Intent Identification Model Training, click Copy ChatGPT Intent & Training Prompt.
3. Paste the prompt into ChatGPT. It generates a single JSON object with 8 Intent Codes (01 to 08) and 400 realistic employee emails (~65% English + ~35% Hinglish, ~50 emails per intent).
4. Paste ChatGPT JSON output into the Stage 2 textarea. Verify the green indicator: Valid JSON Ready.
5. Click Train Stage 2 Intent Model Now (.pkl + TF-IDF Vectorizer). The backend trains the LinearSVC + TF-IDF model, writes the .pkl files, and hot-reloads the model immediately.

STEP 4: Live Email Verification (Stage 3 Tester)

1. In Stage 3 - Live Email Verification, enter a test Email Subject and Body (in English or Hinglish).
2. Click Test Stage 1 Department Detection & Stage 2 Intent Prediction.
3. Verify the Stage 1 Detected Department + Confidence %, Stage 2 Predicted Intent Code + Name + SVM Score + Margin, and Routed HR Desk.

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3. CHATGPT TRAINING JSON SCHEMA EXAMPLE

```
{
  "category_code": "PAYROLL",
  "display_name": "Payroll Operations HR",
  "keywords_strong": ["salary slip", "salary not credited", "salary nahi aayi", "payslip chahiye"],
  "keywords_weak": ["payroll", "salary", "payslip", "tds", "pf", "reimbursement"],
  "required_fields": ["employee_id", "date", "description"],
  "default_intent_id": "PA01",
  "intents": {
    "PA01": "Salary Not Credited or Delayed",
    "PA02": "Salary Deduction or Discrepancy",
    "PA03": "Payslip Request or Payslip Issue"
  },
  "training_samples": [
    {"text": "Subject: Salary not credited\nHi HR, EMP1002 here, October salary not credited.", "intent": "PA01"
    {"text": "Subject: Salary abhi tak nahi aayi\nMera ID EMP1008 hai, salary bank me nahi aayi.", "intent": "PA
  ]
}
```